Chapter 5. Factor groups

5.2 Factor groups

$$
\begin{tikzcd}
G \arrow[rr, "f"] \arrow[d, "\pi"'] & & H \\
G/\ker(f) \arrow[urr, dashed, "\tilde{f}"'] & &
\end{tikzcd}
$$

In this video, we will construct a new type of groups called factor groups.

Constructing factor groups

Let G be a group and H a subgroup of G. We can introduce an intuitive operation in the set of all the cosets of H in G, called the *coset operation*, as the coset of a, operated with the coset of b is defined to be the coset of ab. In symbols,

$$Ha \cdot Hb = H(ab)$$

Take $G = S_3$ and $H = \{\varepsilon, (1\ 2)\}$

$$\underline{H(1\ 3)} = \{(1\ 3), (1\ 3\ 2)\} = \underline{H(1\ 3\ 2)}$$

$$H\underline{(2\ 3)} = \{(2\ 3), (1\ 2\ 3)\} = H\underline{(1\ 2\ 3)}$$

$$H(1\ 3) \cdot H(2\ 3) \overset{\text{def}}{=} H(1\ 3)(2\ 3) = H(1\ 3\ 2)$$

$$\parallel \qquad\qquad \parallel \qquad\qquad\qquad\qquad\qquad\qquad \neq$$

$$H(1\ 3\ 2) \cdot H(1\ 2\ 3) \overset{\text{def}}{=} H(\underbrace{(1\ 3\ 2)(1\ 2\ 3)}_{\varepsilon}) = H$$

?

<u>What did it wrong?</u>

THEOREM **5.6.** *Let H be a normal subgroup of a group G. If $Ha = Hc$ and $Hb = Hd$ then $H(ab) = H(cd)$.*

Proof. Suppose that H is a normal subgroup of a group G. Assume that $Ha = Hc$ and $Hb = Hd$. We need to show that $H(ab) = H(cd)$. By the coset properties, it is enough to show that $ab \in H(cd)$.

$Ha = Hc \Rightarrow a \in Hc \Rightarrow a = h_1 c$ for some $h_1 \in H$

$Hb = Hd \Rightarrow b \in Hd \Rightarrow b = h_2 d$ for some $h_2 \in H$.

$ab = (h_1 c)(h_2 d) = h_1 (c h_2) d$

$ch_2 \in cH = Hc \Rightarrow ch_2 = \underline{h_3 c}$ for some $h_3 \in H$

H normal

$ab = h_1 (ch_2) d = h_1 (h_3 c) d = (h_1 h_3)(cd) \in H(cd)$ ∎

Let G be a group and H a normal subgroup of G. Let G/H be the **set consisting of all the cosets of H in G**. Put it this way, if Ha, Hb, $Hc, \ldots$ are the cosets of H in G, then $G/H = \{Ha, Hb, Hc, \ldots\}$ or, more generally,

$$G/H = \{Ha, Hb, Hc, \ldots\} = \{Hg \mid g \in G\}.$$

We have just proved that the coset operation is indeed an operation on G/H. In our next result, we prove that G/H is a group under coset operation.

THEOREM **5.7.** *Let G be a group and H a* **normal subgroup** *of G. Then G/H is a group under coset operation: $Ha \cdot Hb = H(ab)$*

Proof.

•) Coset operation is associative:

$$(Ha \cdot Hb) \cdot Hc = H(ab) \cdot Hc = H(\overbrace{(ab)}^{\text{operation in } G}c)$$

$$= H(a(bc)) = Ha \cdot H(bc) = Ha \cdot (Hb \cdot Hc)$$

•) The identity element is $H = H1$

$$Ha \cdot H = H(a \cdot 1) = Ha = H(1 \cdot a) = H1 \cdot Ha$$

•) The inverse of Ha is Ha^{-1}:

$$(Ha)(Ha^{-1}) = H(aa^{-1}) = H1 = H$$

$$(Ha^{-1})(Ha) = \ldots = H \quad \blacksquare$$

Recall that $|G : H|$ denotes the number of distinct cosets of H in G, and it is called the **index of H in G.** The group G/H is said to be the **factor group**, or **quotient group** of G by H. Note that $|G/H| = |G : H|$.

An important map is the **coset map** (also called **canonical projection**) $\pi : G \to G/H$ that assigns each element of G to its coset in H, that is, $\pi(a) = Ha$. It is straightforward to check that the coset map is a **group epimorphism** with kernel $\ker \pi = H$. $\hookrightarrow$ Exercise

Example 5.8. Let $\mathbb{Z}$ be the additive cyclic group of the integers, and let $\langle 6 \rangle = 6\mathbb{Z}$ be the cyclic subgroup of $\mathbb{Z}$ consisting of all the multiples of 6.

Notice that $\mathbb{Z}$ is an abelian group, so $6\mathbb{Z}$ is a normal subgroup of $\mathbb{Z}$. So, $\mathbb{Z}/6\mathbb{Z}$ (the cosets of $6\mathbb{Z}$ in $\mathbb{Z}$) is a group.

$$6\mathbb{Z} + 0 = \{\dots, -18, -12, -6, 0, 6, 12, 18, \dots\}$$

$$6\mathbb{Z} + 1 = \{\dots, -17, -11, -5, 1, 7, 13, 19, \dots\}$$

$$6\mathbb{Z} + 2 = \{\dots, -16, -10, -4, 2, 8, 14, 20, \dots\}$$

$$6\mathbb{Z} + 3 = \{\dots, -15, -9, -3, 3, 9, 15, 21, \dots\}$$

$$6\mathbb{Z} + 4 = \{\dots, -14, -8, -2, 4, 10, 16, 22, \dots\}$$

$$6\mathbb{Z} + 5 = \{\dots, -13, -7, -1, 5, 11, 17, 23, \dots\}$$

$(H+a)+(H+b) = H+(a+b)$

$$(6\mathbb{Z}+a)+(6\mathbb{Z}+b) = 6\mathbb{Z}+(a+b).$$

To ease our notation, we write the coset in the following shorter form:

$$\bar{0} = 6\mathbb{Z}+0 \quad \bar{1} = 6\mathbb{Z}+1 \quad \bar{2} = 6\mathbb{Z}+2 \quad \bar{3} = 6\mathbb{Z}+3 \quad \bar{4} = 6\mathbb{Z}+4 \quad \bar{5} = 6\mathbb{Z}+5$$

For example:

$\bar{3}+\bar{4} = (6\mathbb{Z}+3)+(6\mathbb{Z}+4) = 6\mathbb{Z}+(3+4)$

$= 6\mathbb{Z}+7 = 6\mathbb{Z}+1$ because

$7-1 = 6 \in 6\mathbb{Z}$

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$
$\bar{5}$	$\bar{5}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$

$\mathbb{Z}_6$

$Ha = Hb \iff ab^{-1} \in H$

$H+a = H+b \iff a-b \in H$

We have that $\mathbb{Z}/_{6\mathbb{Z}} \cong \mathbb{Z}_6$.

In general:

$$\mathbb{Z}/_{n\mathbb{Z}} \cong \mathbb{Z}_n$$